Geometry, Triangles & Right-Triangle Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, the legs are 5 and 12. What is the hypotenuse?

2 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A right triangle has legs of length 6 and 8. What is the length of the hypotenuse?

3 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a triangle with base 7 and height 10?

4 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, the side opposite angle θ has length 8, the side adjacent to θ has length 15, and the hypotenuse has length 17. What is $\cos(\theta)$?

5 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A right triangle has legs of length 9 and 40. What is the length of the hypotenuse?

6 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a rectangle with length 6 and width 5?

7 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A pole casts a shadow 3 feet long. The distance from the top of the pole to the tip of the shadow is 5 feet. How tall is the pole?

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A circle has equation $(x-2)^2+(y+1)^2=16$. What is the radius?

9 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A right triangle has legs of length 5 and 12. What is the length of the hypotenuse?

10 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the coordinate plane, point A is at $(1, 2)$ and point B is at $(5, 5)$. What is AB?

© ClipSAT — clipsat.org | All Rights Reserved

11 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a rectangle with length 10 and width 3?

12 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

Triangle ABC is similar to triangle DEF. Side $AB = 10$ corresponds to side $DE = 40$, and side $AC = 5$ corresponds to side DF. What is the length of DF?

13 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

Triangle ABC is similar to triangle DEF. Side $AB = 9$ corresponds to side $DE = 45$, and side $AC = 5$ corresponds to side DF. What is the length of DF?

14 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A pole casts a shadow 15 feet long. The distance from the top of the pole to the tip of the shadow is 25 feet. How tall is the pole?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, $\sin\theta = \frac{3}{5}$. What is $\cos\theta$?

16 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a triangle with base 15 and height 12?

17 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A right triangle has legs of length 3 and 4. What is the length of the hypotenuse?

18 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A rectangle has perimeter 36 cm and length = twice width. What is the area in cm²?

19 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A ladder 29 feet long leans against a wall. The base of the ladder is 20 feet from the wall. How high up the wall does the ladder reach?

20 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

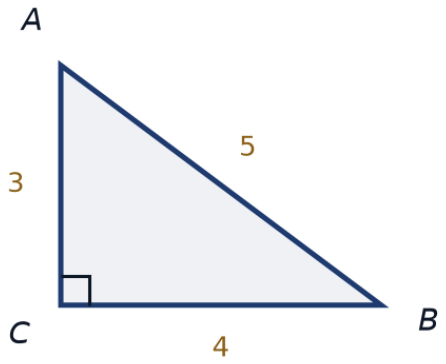
In a $45^\circ - 45^\circ - 90^\circ$ triangle, the hypotenuse is $10\sqrt{2}$. What is the length of each leg?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the right triangle shown, what is the value of $\cos B$?



a. $\frac{4}{5}$

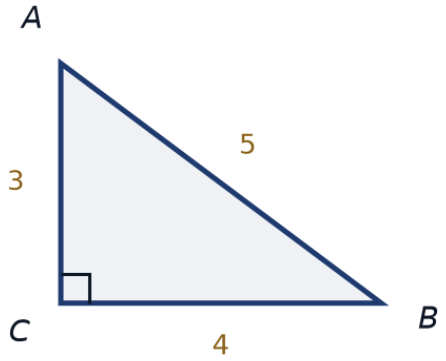
b. $\frac{3}{5}$

c. $\frac{4}{3}$

d. $\frac{5}{3}$

22 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the right triangle shown, what is the value of $\sin A$?



a. $\frac{5}{4}$

b. $\frac{4}{5}$

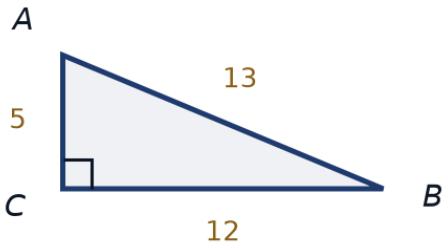
c. $\frac{3}{5}$

d. $\frac{3}{4}$

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the right triangle shown, what is the value of $\cos A$?



a. $\frac{12}{13}$

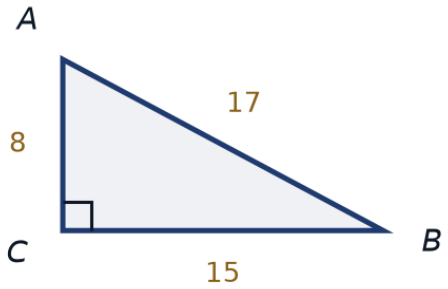
b. $\frac{5}{12}$

c. $\frac{13}{12}$

d. $\frac{5}{13}$

24 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the right triangle shown, what is the value of $\sin A$?



a. $\frac{15}{17}$

b. $\frac{8}{17}$

c. $\frac{8}{15}$

d. $\frac{17}{15}$

25 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, $\sin\theta = \frac{5}{13}$. What is $\cos\theta$?

a. $\frac{5}{12}$

b. $\frac{12}{13}$

c. $\frac{13}{12}$

d. $\frac{5}{13}$

26 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

Two parallel lines are cut by a transversal. One co-interior angle measures 65°. What is the other?

a. 65°

b. 115°

c. 125°

d. 25°

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a triangle, two of the angles measure 21° and 59°. What is the measure of the third angle?

a. 100°

b. 105°

c. 108°

d. 104°

28 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A right triangle has a hypotenuse of length 26 and one leg of length 10. What is the length of the other leg?

a. 30

b. 31

c. 24

d. 22

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a triangle, two of the angles measure 38° and 74°. What is the measure of the third angle?

a. 64°

b. 68°

c. 66°

d. 59°

30 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a circle with radius 8? (Express your answer in terms of π.)

a. 128π

b. 81π

c. 64π

d. 16π

31 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the circumference of a circle with a diameter of 14 inches?

a. 7π in

b. 14π in

c. 28π in

d. 49π in

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the radius of the circle with equation $(x-3)^2+(y+5)^2=49$?

a. 5

b. 7

c. 14

d. 49

33 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, the legs have lengths 9 and 12. What is the area of the triangle?

a. 27

b. 54

c. 72

d. 108

34 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A square has an area of $64\ \text{m}^2$. What is its perimeter?

a. 16 m

b. 24 m

c. 32 m

d. 48 m

35 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the volume of a cylinder with radius 3 cm and height 10 cm?

a. $30\pi\ \text{cm}^3$

b. $60\pi\ \text{cm}^3$

c. $90\pi\ \text{cm}^3$

d. $120\pi\ \text{cm}^3$

36 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the surface area of a cube with edge length 4 cm?

a. 64 cm²

b. 96 cm²

c. 128 cm²

d. 144 cm²

37 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a parallelogram with base 12 cm and height 5 cm?

a. 30 cm²

b. 60 cm²

c. 90 cm²

d. 120 cm²

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, the side opposite angle θ has length 20, the side adjacent to θ has length 21, and the hypotenuse has length 29. What is $\tan(\theta)$?

a. $121/105$

b. $20/21$

c. $13/21$

d. $59/84$

39 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a circle with a radius of 6 cm?

a. 12π cm^2

b. 24π cm^2

c. 36π cm^2

d. 72π cm^2

40 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A rectangle has a width of 7 cm and a diagonal of 25 cm. What is the length of the rectangle?

a. 20 cm

b. 22 cm

c. 24 cm

d. 26 cm

41 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the center of the circle given by $(x + 2)^2 + (y - 8)^2 = 25$?

a. $(2, -8)$

b. $(-2, 8)$

c. $(-2, -8)$

d. $(2, 8)$

42 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the volume of a cube with edge length 5 cm?

a. 25 cm³

b. 75 cm³

c. 100 cm³

d. 125 cm³

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, acute angle A measures 30°. What is $\sin(30^\circ)$?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$

d. 1

44 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle with legs 5 and 12, what is $\tan(\theta)$ if θ is opposite the leg of length 5?

a. $\frac{5}{12}$

b. $\frac{12}{5}$

c. $\frac{5}{13}$

d. $\frac{12}{13}$

45 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

The diameter of a circle is 18. What is the area of the circle?

a. 36π

b. 81π

c. 144π

d. 324π

© ClipSAT — clipsat.org | All Rights Reserved